

chapter-01

July 10, 2026

1 Infinite Series, Power Series

```
[ ]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import math
from IPython import display
from IPython.display import Markdown
from scipy.special import factorial
from scipy import stats
from scipy import constants

# Below are some adjustments:
# from IPython.display import HTML
# HTML('<style>img[alt="skan"] { filter: invert(100%); }</style>')
# plt.style.use('dark_background')
```

1.1 Problems, Section 2

In the following problems, find the limit of the given sequence as $n \rightarrow \infty$.

1.1.1 2.6

$$\lim_{n \rightarrow \infty} \frac{n^n}{n!}$$

```
[ ]: # 1. Simulation setup
# We keep the range small (1 to 10) because n^n grows insanely fast.
n_max: float = 10.0
step: float = 0.5
start: float = 1.0
n_values = np.arange(start, n_max, step)
custom_mk = 2

numerators = n_values**n_values
denominators = factorial(n_values)
results = numerators / denominators
```

```

# 3. Plotting the chart
plt.figure(figsize=(10, 6))

# Draw the main line
plt.plot(n_values, results, color='red', linestyle=':', marker='o',
        ↳markevery=custom_mk, label=r'values  $\frac{n^n}{n!}$ ')

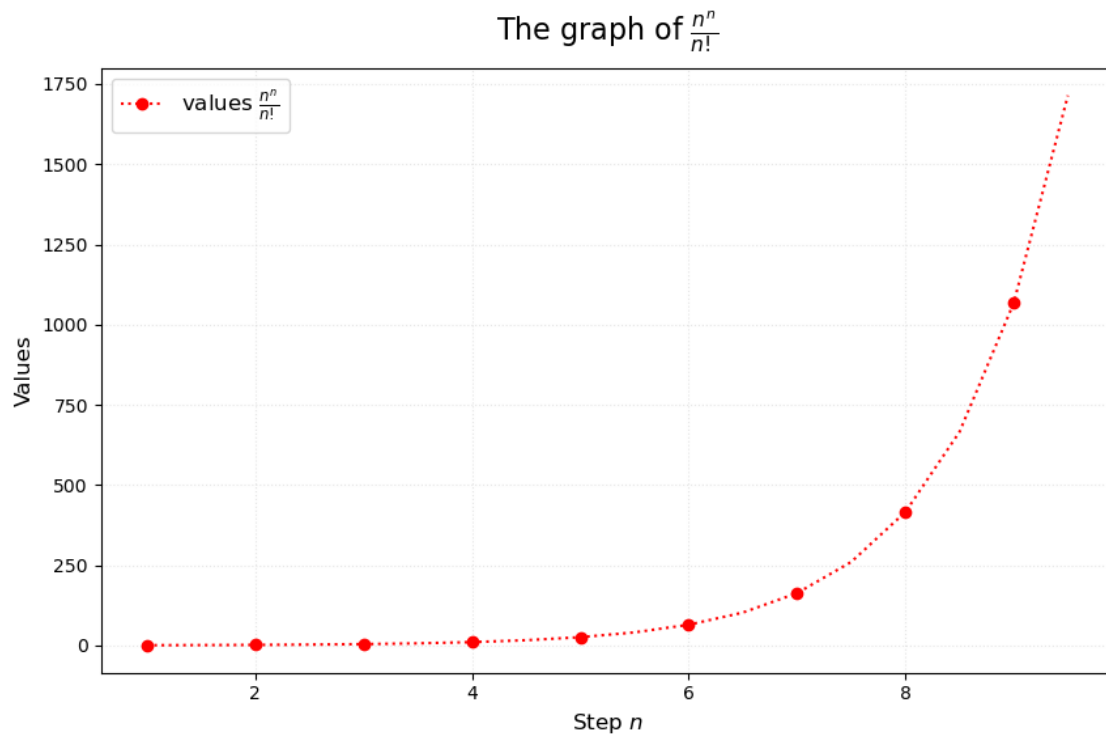
# You may ask "why is this code excluded?"
# Let's have a look at the table of Values below
# It's reasonable not to include numerator and denominator because of the rapid
↳growth of those values
#plt.plot(n_values, numerators, color='blue', linestyle='-', marker='s',
↳markevery=5, label=r'numerator  $n^n$ ')
#plt.plot(n_values, denominators, color='green', linestyle='--', marker='^',
↳markevery=5, label=r'denominator  $n!$ ')

# Aesthetics and labels
plt.title(r'The graph of  $\frac{n^n}{n!}$ ', fontsize=16, pad=15)
plt.xlabel('Step  $n$ ', fontsize=12)
plt.ylabel('Values', fontsize=12)
plt.grid(True, linestyle=':', alpha=0.3)
plt.legend(fontsize=12)

# Render the plot
plt.show()

table_data = {
    r"Step  $n$ ": n_values,
    r"Numerator  $n^n$ ": numerators,
    r"denominator  $n!$ ": denominators,
    r"Value  $\frac{n^n}{n!}$ ": results
}
df = pd.DataFrame(table_data)
Markdown(df.to_markdown(index=False))

```



[]:

Step n	Numerator n^n	denominator $n!$	Value $\frac{n^n}{n!}$
1	1	1	1
1.5	1.83712	1.32934	1.38198
2	4	2	2
2.5	9.88212	3.32335	2.97354
3	27	6	4.5
3.5	80.2118	11.6317	6.89595
4	256	24	10.6667
4.5	869.874	52.3428	16.6188
5	3125	120	26.0417
5.5	11803.1	287.885	40.9992
6	46656	720	64.8
6.5	192281	1871.25	102.755
7	823543	5040	163.401
7.5	3.65561e+06	14034.4	260.475
8	1.67772e+07	40320	416.102
8.5	7.9444e+07	119292	665.96
9	3.8742e+08	362880	1067.63
9.5	1.94256e+09	1.13328e+06	1714.11

$$\lim_{n \rightarrow \infty} \frac{n^n}{n!} = \infty$$

1.1.2 2.7

$$\lim_{n \rightarrow \infty} \left((1 + n^2)^{\frac{1}{\ln n}} \right)$$

```
[59]: def compute():
    n_max = 20
    n_start = 2.0
    n_step = 0.1
    n_values = np.arange(n_start, n_max, n_step)
    results = (1 + n_values**2)**(1 / np.log(n_values))
    powers = 1 / np.log(n_values)
    bases = 1 + n_values**n_values

    plt.figure(figsize=(10,6))

    plt.plot(n_values, results, color='red', linestyle="--", marker='o',
    ↳markevery=10, label=r'Values  $(1+n^2)^{\frac{1}{\ln n}}$ ')

    euler_value = np.exp(2)

    plt.axhline(y=euler_value, color='blue', linestyle='--',linewidth=2,
    ↳label=r'$e^2$')
    plt.title(r"The graph of  $(1+n^2)^{\frac{1}{\ln n}}$ ", fontsize=16, pad=15)
    plt.xlabel('$n$', fontsize=14)
    plt.ylabel('Values', fontsize=14)
    plt.grid(True, linestyle=':', alpha=0.5)
    plt.legend(fontsize=14, loc='upper right')

    plt.show()

    table_data = {
        r"Step  $n$ ": n_values,
        r"Base  $1 + n^2$ ": bases,
        r"Power  $\frac{1}{\ln n}$ ": powers,
        r"Values  $(1+n^2)^{\frac{1}{\ln n}}$ ": np.round(results, 4)
```

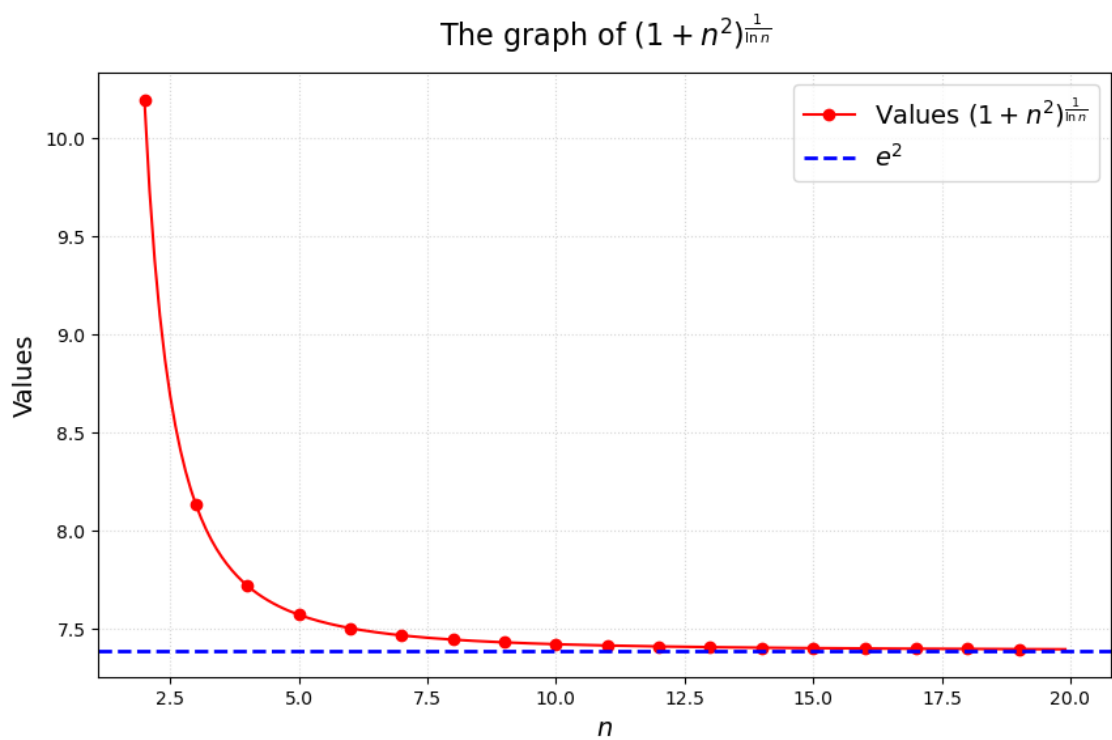
```

}

df = pd.DataFrame(table_data)
df_sliced = df.iloc[::20]
return Markdown(df_sliced.to_markdown(index=False))

compute()

```



[59] :

Step n	Base $1 + n^2$	Power $\frac{1}{\ln n}$	Values $(1 + n^2)^{\frac{1}{\ln n}}$
2	5	1.4427	10.1953
4	257	0.721348	7.7194
6	46657	0.558111	7.5029
8	1.67772e+07	0.480898	7.4444
10	1e+10	0.434294	7.4211
12	8.9161e+12	0.40243	7.4097
14	1.1112e+16	0.378923	7.4033
16	1.84467e+19	0.360674	7.3995
18	3.93464e+22	0.345976	7.3969

$$\begin{aligned}
 2.4 \quad \lim_{n \rightarrow \infty} (1+n^2)^{\frac{1}{\ln n}} &= \left[\infty^0 \right] \\
 (1+n^2)^{\frac{1}{\ln n}} &= e^{\ln \left[(1+n^2)^{\frac{1}{\ln n}} \right]} = e^{\frac{\ln(1+n^2)}{\ln n}} = e^{\frac{\ln n^2 + \ln \left[1 + \frac{1}{n^2} \right]}{\ln n}} \\
 \lim_{n \rightarrow \infty} (1+n^2)^{\frac{1}{\ln n}} &= \lim_{n \rightarrow \infty} e^{\frac{\ln n^2 + \ln \left[1 + \frac{1}{n^2} \right]}{\ln n}} = \lim_{n \rightarrow \infty} e^{\frac{2 \ln n + \ln 1}{\ln n}} = \underline{e^2}
 \end{aligned}$$

1.1.3 2.8

$$\lim_{n \rightarrow \infty} \frac{(n!)^2}{(2n)!}$$

```
[ ]: def compute():
    max_n = 1024
    n_values = list(range(2, max_n+1))
    results = []

    for n in n_values:
        numerator = math.factorial(n)**2
        denominator = math.factorial(2*n)
        value = numerator / denominator
        results.append(value)

    for i in [2,4,8,16,32,64,128,256,512, 1023]:
        print(f"For {i} the value is approximately {results[i-1]:.4f}")

compute()
```

```
For 2 the value is approximately 0.0500
For 4 the value is approximately 0.0040
For 8 the value is approximately 0.0000
For 16 the value is approximately 0.0000
For 32 the value is approximately 0.0000
For 64 the value is approximately 0.0000
For 128 the value is approximately 0.0000
For 256 the value is approximately 0.0000
For 512 the value is approximately 0.0000
For 1023 the value is approximately 0.0000
```

2.8

$$\lim_{n \rightarrow \infty} \frac{(n!)^2}{(2n)!}$$

$$(n!)^2 = (1 \cdot 2 \cdot 3 \cdot \dots \cdot n)(1 \cdot 2 \cdot 3 \cdot \dots \cdot n)$$

$$(2n)! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot n(n+1) \cdot (n+2) \cdot \dots \cdot 2n$$

$$a_n = \frac{n! \cdot n!}{n! \cdot (n+1)(n+2) \cdot \dots \cdot 2n} = \frac{n!}{(n+1)(n+2) \cdot \dots \cdot 2n} =$$

$$= \frac{1}{n+1} \cdot \frac{2}{n+2} \cdot \frac{3}{n+3} \cdot \dots \cdot \frac{n}{n+n}$$

$$a_n \leq \left(\frac{1}{2}\right)^n \Rightarrow \lim_{n \rightarrow \infty} \frac{(n!)^2}{(2n)!} = 0$$

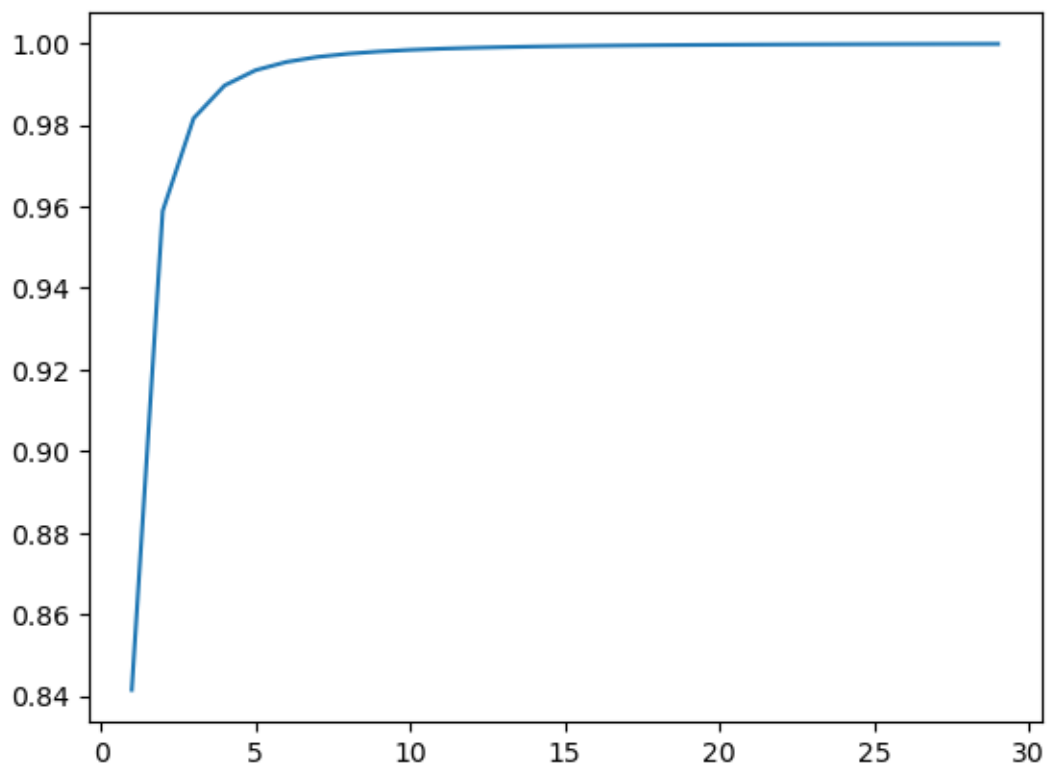
1.1.4 2.9

$$\lim_{n \rightarrow \infty} n \sin\left(\frac{1}{n}\right)$$

```
[6]: def compute():
    n_max = 30
    n_values = list(range(1, n_max))
    results = []

    for n in n_values:
        value = n * math.sin(1/n)
        results.append(value)

    plt.plot(n_values, results)
    plt.show()
compute()
```



2.9

$$\lim_{n \rightarrow \infty} n \sin \frac{1}{n} = [0 \cdot \infty]$$

$\nearrow \infty$
 $\searrow 0$

$x = \frac{1}{n} \Rightarrow n = \frac{1}{x}$
 As $n \rightarrow \infty$, $\frac{1}{x}$ approaches 0

$$\lim_{n \rightarrow \infty} n \sin \frac{1}{n} = \lim_{x \rightarrow 0} \frac{1}{x} \sin x = \lim_{x \rightarrow 0} \frac{\sin x}{x} =$$

$$= \lim_{n \rightarrow \infty} 1 \cdot \frac{\sin x}{x} = 1$$

1.2 Problems, Section 4

For the following series, write formulas for the sequences a_n , S_n , and R_n , and find the limits of the sequences as $n \rightarrow \infty$ (if the limits exist).

1.2.1 4.1

$$\sum_{n=1}^{\infty} \frac{1}{2^n}$$

h.1.

$$\sum_{n=1}^{\infty} \frac{1}{2^n}$$

$$a_n = \frac{1}{2^n}$$

$$S_n = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n}$$

$$a_1 = \frac{1}{2} \quad \left. \begin{array}{l} r = \frac{1}{2} \end{array} \right\} S_n = \frac{a_1 \cdot (1 - r^n)}{1 - r} = \frac{\frac{1}{2} [1 - (\frac{1}{2})^n]}{1 - \frac{1}{2}}$$

$$S_n = \frac{\frac{1}{2} - (\frac{1}{2})^{n+1}}{\frac{1}{2}} = 1 - (\frac{1}{2})^n$$

$$R_n = S - S_n = \lim_{n \rightarrow \infty} S_n - S_n =$$

$$= 1 - 1 + (\frac{1}{2})^n = (\frac{1}{2})^n$$

1.2.2 4.2

$$\sum_{n=0}^{\infty} \frac{1}{5^n}$$

4.2

$$\sum_0^{\infty} \frac{1}{5^n} = 1 + \frac{1}{5} + \frac{1}{25} + \dots$$

I. $a_n = \frac{1}{5^{n-1}}$ *usually, geometric progression starts with $n=1$, but in this case $a_1=1$, so we need to subtract 1 from n*

II. $a_1 = 1$ $r = \frac{1}{5}$ $\Rightarrow S_n = 1 \cdot \frac{[1 - (\frac{1}{5})^n]}{1 - \frac{1}{5}} = \frac{1 - (\frac{1}{5})^n}{\frac{4}{5}} = \frac{5 - (\frac{1}{5})^{n-1}}{4} = \frac{5}{4} [1 - (\frac{1}{5})^n]$

III. $\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left[\frac{5}{4} [1 - (\frac{1}{5})^n] \right] = \left[\text{as } n \rightarrow \infty, (\frac{1}{5})^n \rightarrow 0 \right] = \frac{5}{4}$

IV. $R_n = S - S_n = \frac{5}{4} - \frac{5}{4} [1 - (\frac{1}{5})^n] = \frac{5}{4} - \frac{5}{4} + \frac{5}{4} (\frac{1}{5})^n = \frac{1}{4} \cdot (\frac{1}{5})^{n-1}$

V. $\lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} \frac{1}{4} \cdot (\frac{1}{5})^{n-1} = \left[\text{as } n \rightarrow \infty, (\frac{1}{5})^{n-1} \rightarrow 0 \right] = 0$

1.2.3 4.3

$$1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} \dots$$

4.3

$$1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16}$$

I. $a_n = \frac{1}{(-2)^{n-1}}$

II. $a_1 = 1$ $r = -\frac{1}{2}$ $\Rightarrow S_n = 1 \cdot \frac{1 - (-\frac{1}{2})^n}{1 - (-\frac{1}{2})} = \frac{1 - (-\frac{1}{2})^n}{\frac{3}{2}} = \frac{2}{3} [1 - (-\frac{1}{2})^n]$

III. $\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{2}{3} [1 - (-\frac{1}{2})^n] = \left[\text{as } n \rightarrow \infty, (-\frac{1}{2})^n \rightarrow 0 \right] = \frac{2}{3}$

IV. $R_n = S - S_n = \frac{2}{3} - \frac{2}{3} [1 - (-\frac{1}{2})^n] = \frac{2}{3} (-\frac{1}{2})^n$

V. $\lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} \frac{2}{3} (-\frac{1}{2})^n = \left[\text{as } n \rightarrow \infty, (-\frac{1}{2})^n \rightarrow 0 \right] = 0$

1.2.4 4.4

$$\sum_1^{\infty} e^{-n \ln 3}$$

$$\begin{aligned}
 & \sum_{n=1}^{\infty} e^{-n \ln 3} \quad \nearrow \ln 3 = \log_e 3, \quad x = e^{\ln x} \\
 & e^{-n \ln 3} = (e^{-\ln 3})^n = \left(\frac{1}{3}\right)^n \\
 \text{I} \quad & a_n = \left(\frac{1}{3}\right)^n \\
 \text{II} \quad & a_1 = \frac{1}{3} \quad \left. \begin{array}{l} r = \frac{1}{3} \end{array} \right\} S_n = \frac{1}{3} \cdot \frac{1 - \left(\frac{1}{3}\right)^n}{1 - \frac{1}{3}} = \frac{1}{2} \cdot \cancel{2} \cdot \left[1 - \left(\frac{1}{3}\right)^n\right] = \frac{1}{2} \left[1 - \left(\frac{1}{3}\right)^n\right] \\
 \text{III} \quad & \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{1}{2} \left[1 - \left(\frac{1}{3}\right)^n\right] = \left[\text{as } n \rightarrow \infty, \left(\frac{1}{3}\right)^n \rightarrow 0 \right] = \frac{1}{2} \\
 \text{IV} \quad & R_n = S - S_n = \frac{1}{2} - \frac{1}{2} \left[1 - \left(\frac{1}{3}\right)^n\right] = \frac{1}{2} \left(\frac{1}{3}\right)^n \\
 \text{V} \quad & \lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} \frac{1}{2} \left(\frac{1}{3}\right)^n = \left[\text{as } n \rightarrow \infty, \left(\frac{1}{3}\right)^n \rightarrow 0 \right] = 0
 \end{aligned}$$

1.2.5 4.5

$$\sum_0^{\infty} e^{2n \ln \sin\left(\frac{\pi}{3}\right)}$$

$$4.5 \quad \sum_0^{\infty} e^{2n \ln \sin \frac{\pi}{3}}$$

$$e^{2n \ln \sin \frac{\pi}{3}} = e^{2n \ln \frac{\sqrt{3}}{2}} = \left[e^{2 \ln \frac{\sqrt{3}}{2}} \right]^n = \left[e^{\ln \frac{3}{4}} \right]^n = \left(\frac{3}{4} \right)^n$$

I $a_n = \left(\frac{3}{4} \right)^{n-1}$

II $a_1 = 1$
 $r = \frac{3}{4} \quad \left. \vphantom{\begin{matrix} a_1 = 1 \\ r = \frac{3}{4} \end{matrix}} \right\} S_n = 1 \cdot \frac{1 - \left(\frac{3}{4} \right)^n}{1 - \frac{3}{4}} = \frac{1 - \left(\frac{3}{4} \right)^n}{\frac{1}{4}} = 4 \left[1 - \left(\frac{3}{4} \right)^n \right]$

III $\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} 4 \left[1 - \left(\frac{3}{4} \right)^n \right] = \left[\text{as } n \rightarrow \infty, \left(\frac{3}{4} \right)^n \rightarrow 0 \right] = 4$

IV $R_n = S - S_n = 4 - 4 \left[1 - \left(\frac{3}{4} \right)^n \right] = 4 \left(\frac{3}{4} \right)^n$

V $\lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} 4 \left(\frac{3}{4} \right)^n = \left[\text{as } n \rightarrow \infty, \left(\frac{3}{4} \right)^n \rightarrow 0 \right] = 0$

1.2.6 4.6

$$\sum_1^{\infty} \frac{1}{n(n+1)}$$

h. 6

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \sum_{n=1}^{\infty} \left[\frac{1}{n} - \frac{1}{n+1} \right]$$

$$\left. \begin{array}{l} a_1 = 1 - \frac{1}{2} = \frac{1}{2} \\ a_2 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6} \\ a_3 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12} \\ a_n = \frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)} \\ a_5 = \frac{1}{5} - \frac{1}{6} = \frac{1}{30} \end{array} \right\} \begin{array}{l} I \ a_n = \frac{1}{n(n+1)} \\ \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n(n+1)} = 0 \end{array}$$

$$II \quad \sum_{n=1}^{\infty} \left[\frac{1}{n} - \frac{1}{n+1} \right] = 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{n} - \frac{1}{n+1} = 1 - \frac{1}{n+1} \quad \text{as } n \rightarrow \infty$$

$$S_n = 1 - \frac{1}{n+1}$$

$$III \quad \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left[1 - \frac{1}{n+1} \right] = \left[\text{as } n \rightarrow \infty, \frac{1}{n+1} \rightarrow 0 \right] = 1$$

$$IV \quad R_n = S - S_n = 1 - 1 + \frac{1}{n+1} = \frac{1}{n+1}$$

$$V \quad \lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} \frac{1}{n+1} = 0$$

1.2.7 4.7

$$\frac{3}{1 \cdot 2} - \frac{5}{2 \cdot 3} + \frac{7}{3 \cdot 4} - \frac{9}{4 \cdot 5} + \dots$$

h. 7

$$\frac{3}{1 \cdot 2} - \frac{5}{2 \cdot 3} + \frac{7}{3 \cdot 4} - \frac{9}{4 \cdot 5} + \dots =$$

$$\text{I } a_n = \frac{(-1)^{n-1} \cdot (2n+1)}{n(n+1)}, \quad \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{(-1)^{n-1} \cdot (2n+1)}{n(n+1)} = 0$$

$$\text{II } S_n = S_b - S_c$$

$$S_b = \frac{3}{1 \cdot 2} + \frac{7}{3 \cdot 4} + \frac{11}{5 \cdot 6} + \dots = \frac{3}{1} - \frac{3}{2} + \frac{7}{3} - \frac{7}{4} + \frac{11}{5} - \frac{11}{6} + \dots$$

$$S_c = \frac{5}{2} - \frac{5}{3} + \frac{9}{4} - \frac{9}{5} + \frac{13}{6} - \frac{13}{7} + \dots$$

$$S_n = 3 - \frac{3}{2} - \frac{5}{2} + \frac{7}{3} + \frac{5}{3} - \frac{7}{4} - \frac{9}{4} + \frac{11}{5} + \frac{9}{5} - \frac{11}{6} - \frac{13}{6} + \dots$$

$$S_n = 3 - \frac{8}{2} + \frac{12}{3} - \frac{16}{4} + \frac{20}{5} - \frac{24}{6} + \dots$$

$$S_n = 3 - 4 + 4 - 4 + 4 - 4 + 4 - 4 + \dots + \frac{(-1)^{n-1} \cdot (2n+1)}{n(n+1)}$$

$$S_n = 3 + (-1)^{n-1} \cdot \frac{(2n+1)}{n(n+1)}$$

$$\text{III } \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left[3 + (-1)^n \frac{2n+1}{n(n+1)} \right] = \left[\text{as } n \rightarrow \infty, \frac{2n+1}{n(n+1)} \rightarrow 0 \right] = 3$$

$$\text{IV } R_n = S - S_n = 3 - 3 + \frac{(-1)^n (2n+1)}{n(n+1)} = \frac{(-1)^n (2n+1)}{n(n+1)}$$

$$\text{V } \lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} \frac{(-1)^n (2n+1)}{n(n+1)} = \left[\text{as } n \rightarrow \infty, \frac{2n+1}{n(n+1)} \rightarrow 0 \right] = 0$$